

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VIth Semester of BSc (BT/BC/MI) Examination April-May 2019

Course code & Name: BE 604 FUNDAMENTALS OF OMICS TECHNOLOGIES

Date: 04.05.2019 Day: Saturday Time: 1.30 PM To 1.50PM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

10

1. Chose the **odd** word from the following

(a) Phosphorylase

(b) Apyrase

(c) Sulphurylase

(d) Luciferase

2. The relationship of any two characters that have descended, usually through divergence, from a common ancestral character.

(a) Analogous

(b) Homologous

(c) Heterologous

(d) Genologous

3. All the RNA molecules transcribed from a genome

(a) Transcriptome

(b) Transcript

(c) Metagenmics

(d) mRNA

4. Chose the **odd** word from the following
- | | | |
|-------------------|--------------|-------|
| (a) Ion source | (b) Adaptor | _____ |
| (c) Mass analyzer | (d) Detector | _____ |
5. The full form of HGP is _____
- | | | |
|--------------------------|------------------------------------|-------|
| (a) Human genome project | (b) Human Growth project | _____ |
| (c) Human gene project | (d) Human Genomics and Proteiomics | _____ |
6. Emulsion PCR
- | | | |
|-----------------|--------------|-------|
| (a) Ion torrent | (b) 454 | _____ |
| (c) Shotgun | (d) Illumina | _____ |
7. Microarray can be considered as a..... system
- | | | |
|-----------------|-----------------|-------|
| (a) Transparent | (b) Overlapping | _____ |
| (c) Open | (d) Closed | _____ |
8. Method that describes the various approaches that seek to determine the rates of metabolic reactions within a biological entity.
- | | | |
|--------------------|-----------------------------|-------|
| (a) Metagenomics | (b) Functional metagenomics | _____ |
| (c) Metabolic flux | (d) Fluxomics | _____ |
9. Comparative genomics involves
- | | | |
|-----------------------------------|----------------------------------|-------|
| (a) Comparison of Gene location | (b) Comparison of Gene structure | _____ |
| (c) Comparison of Gene characters | (d) All of the above | _____ |
10. Solexia sequencing is.....
- | | | |
|-----------------|--------------|-------|
| (a) Ion torrent | (b) 454 | _____ |
| (c) Shotgun | (d) Illumina | _____ |

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VIth Semester of BSc (BT/BC/MI) Examination April-May 2019

Course code & Name: BE 604 FUNDAMENTALS OF OMICS TECHNOLOGIES

Date: 04.05.2019 Day: Saturday Time: 1.50 PM To 3.30PM Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
 2. Make suitable assumptions and draw neat figures wherever required.
 3. Use of non-programmable calculator is allowed.
 4. Show necessary calculations.
-

SECTION - I

Q - II Answer the following questions as directed 10

A. Answer any two of the following 4

- 1 Differentiate between Orthologues and Paralogues
- 2 Define the following: (a) Proteome (b) Metabolome
- 3 What is hierarchical shotgun sequencing?

B. Answer any two of the following 6

- 1 Describe the applications of Mass spectrophotometry
 - 2 What are the basic steps involved in SAGE technology?
 - 3 Describe Sanger sequencing and its advantage.
-

SECTION - II

Q-III Answer any three of the following questions 15

- 1 Describe the 454 pyrosequencing method
- 2 What is Pharmacogenomics? How is it helpful?
- 3 Write a short note on Genomics and its types.
- 4 Describe metabolomics in detail.